

深圳市特深电气有限公司

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Knee Coil Service Manual

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ Това упътване за работа е налично само на английски език.

(BG)

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告

(ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告

(ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告

(ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE

(HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimajte li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS
(FI)

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvit - tavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaar - atilanteen vuoksi.

ATTENTION
(FR)

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG
(DE)

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst - technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΪΗ
(EL)

Σο παπτόν εγσειπιτίδιο ζέπβιρ διαηίθειαι μόνο ζηα αγγλικά.

- Εάν ο ηεσνικόρ ζέπβιρ ενόρ πελάηη απαιηεί ηο παπτόν εγσειπιτίδιο ζε γλώζζα εκηόρ ηων αγγλικών, αποηελεί εςθύνη ηος πελάηη να παπέσει ηιρ ζηηπεζίερ μεηάθηπαζηρ.
- Μην επισειπήζεηε ηηνν εκηέλεζη επγαζιών ζέπβιρ ζηον εξοπιζμό αν δεν έσειε ζςμβοςλεζηεί και καηανοήζει ηο παπτόν εγσειπιτίδιο ζέπβιρ.
- Αν δεν πποζέξεηε ηηνν πποειδοποιήζη αζηή, ενδέσειηαι να πποκληθεί ηπασμαηιζμόρ ζηον ηεσνικό ζέπβιρ, ζηο σειπιζηή ή ζηον αζθενή από ηλεκηποπληζία, μησανικούρ ή άλλοςρ κινδύνοςρ.

FIGYELMEZTETÉS
(HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. · Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. · Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. · Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. · Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. · Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. · Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 · サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 · このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 · この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. · 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. · 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. · 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. · Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. · Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. · Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento suža - lojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL
(NO)

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, oper - atøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłu - maczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia me - chanicznego bądź innego.

ATENÇÃO
(PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi - lačke usluge. • Ne pokušavajte da opravite ureĐaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanimarivanje ovog upozorenja može dovesti do povreĐivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING
(SV)

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

OPOZORILO
(SL)

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka za - gotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

DİKKAT
(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Rev	Date	Description
1.0	2022-04-08	Initial release
2.0	2022-11-24	Update label image
2.1	2023-02-09	Update Indications for Use
3.0	2023-02-11	Add A-Port

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Table of Contents

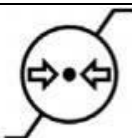
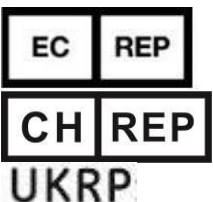
Chapter 1 Coil Introduction	15
1-1 Explanation of symbols	15
1-2 Product Identification and Shipping List	18
1-3 Introduction	20
1-4 Compatibility	20
1-5 Related Documentation	20
1-6 Indications for Use	20
1-7 Qualification for Service Personnel	21
1-8 Warranty Statement	21
1-9 Cautions	21
1-10 Coil Classification	21
1-11 Environmental Requirements	21
Chapter 2 Coil Setup	23
2-1 Knee Coil MCQA Test Setup	23
Chapter 3 Coil Use	27
3-1 Knee Coil Use	27
Chapter 4 Scanning	32
4-1 Autoshim	32
4-2 Positioning	32
4-3 Fat Saturation Techniques	32
4-4 Scanning Protocols	32
Chapter 5 Troubleshooting	33
5-1 Knee Coil Troubleshooting Tips	33
Chapter 6 Maintenance	38
6-1 Quality Assurance	38
6-2 Coil Care	38
6-3 Special Care Requirements	38

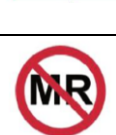
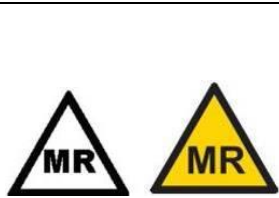
Chapter 7 FRU and Replace procedures	39
7-1 Knee Coil FRUs	39
7-2 Knee Coil Cable Replacement	40

Chapter 1 Coil Introduction

1-1 Explanation of symbols

NO.	Symbol	Description
1		Manufacturer
2		Date of manufacture
3		Catalogue Number
4		Serial Number
5	FIELD	Field Strength
6		(it is a mandatory action to) Refer to the instruction manual/booklet
7		Consult Instructions for Use
8		Type BF Applied Part
9		Class II Equipment
10		RF coil, transmit/receive
11		Dispose of in accordance with local regulation
12	CE	CE marking of conformity' or 'CE marking' means a marking by which a manufacturer indicates that a device is in conformity with the applicable requirements set out in this Regulation and other applicable Union harmonization legislation providing for its affixing.

13		Medical Device
14		Unique Device Identifier
15		Temperature Limits
16		Humidity Limits
17		Atmospheric Pressure Limits
18		Keep dry
19		Fragile, handle with care
20		Keep away from sunlight
21		Front upward
22		The limit of the stack is 3.
23		Ambassadorn
24		British product certification identification.

25		Prescription Device Caution: US Federal law restricts this device to the sale, distribution and use by or on the order of a clinician.
26		The ETL Listed Mark indicates that the product has been tested by Intertek and found in compliance with accepted national standards.
27		Possible pinch hand hazard
28		Warning
29		Burn Hazard
30		Do not cross or loop cables. Arcing and patient burns could result.
31		MR Safe: An item that poses no known hazards resulting from exposure to any MRI environment. MR Safe items are composed of materials that are electrically nonconductive, nonmetallic, and nonmagnetic.
32		MR Unsafe: An item, which poses unacceptable risks to the patient, medical staff, or other persons within the MRI environment.
33		MR Conditional: An item with demonstrated safety in the MRI environment within defined conditions. At a minimum, address the conditions of the static magnetic field, the switched gradient magnetic field and the radiofrequency fields. Additional conditions, including specific configurations of the item, may be required.
34		Non-ionizing Radiation Emission

Shenzhen RF Tech Co.,Ltd

1-2 Product Identification and Shipping List

Identify the Knee Coil by the coil label.

Illustration 1-1: 8ch T/R Knee Coil Label

 Distributed by GE Medical Systems,LLC GE PART NO. 5869241-2, Rev.1		 (01)06938146600102(11)230112(21)00010											
 SHENZHEN RF TECH CO.,LTD 2-F, Bld4, Juhui Industrial Park, Tianliao, Guangming,Shenzhen, P.R.CHINA 518132		8ch T/R Knee Coil <table border="1"> <thead> <tr> <th>REF</th> <th>SN</th> <th>REV.</th> <th>FIELD</th> <th>CR</th> </tr> </thead> <tbody> <tr> <td>10-F31896</td> <td>00010</td> <td>1.0</td> <td>1.5T</td> <td>2023-01-12</td> </tr> </tbody> </table>		REF	SN	REV.	FIELD	CR	10-F31896	00010	1.0	1.5T	2023-01-12
REF	SN	REV.	FIELD	CR									
10-F31896	00010	1.0	1.5T	2023-01-12									
  1639     	 OSMUNDA Medical Technology Service GmbH TRESKOWALLEE 108,10318 BERLIN, GERMANY	 Casus Switzerland GmbH Hinterbergstrasse 49 6312 Steinhausen Switzerland	 Wellkang Ltd 16 Castle St,Dover, CT16 1PW, England,UK										
		Intertek 5002975 Conforms to AAMI STD.ES60601-1 Certified to CSA STD.C22.2 NO.60601-1											

Illustration 1-2: 1.5T HD T/R Knee Array Label

 Distributed by GE Medical Systems,LLC GE PART NO. 5932028-2, Rev.1		 (01)06938146600140(11)230215(21)000001											
 SHENZHEN RF TECH CO.,LTD 2-F, Bld4, Juhui Industrial Park, Tianliao, Guangming,Shenzhen, P.R.CHINA 518132		1.5T HD T/R Knee Array <table border="1"> <thead> <tr> <th>REF</th> <th>SN</th> <th>REV.</th> <th>FIELD</th> <th>CR</th> </tr> </thead> <tbody> <tr> <td>10-F34127</td> <td>00001</td> <td>1.0</td> <td>1.5T</td> <td>2023-02-15</td> </tr> </tbody> </table>		REF	SN	REV.	FIELD	CR	10-F34127	00001	1.0	1.5T	2023-02-15
REF	SN	REV.	FIELD	CR									
10-F34127	00001	1.0	1.5T	2023-02-15									
  1639     	 OSMUNDA Medical Technology Service GmbH TRESKOWALLEE 108,10318 BERLIN, GERMANY	 Casus Switzerland GmbH Hinterbergstrasse 49 6312 Steinhausen Switzerland	 Wellkang Ltd 16 Castle St,Dover, CT16 1PW, England,UK										
		Intertek 5002975 Conforms to AAMI STD.ES60601-1 Certified to CSA STD.C22.2 NO.60601-1											

Illustration 1-3: Knee Coil



This table provides the list of items shipped with the Knee Coil. There contents are applied parts supplied by Shenzhen RF Tech Co.,Ltd.

Illustration 1-4: Applied parts

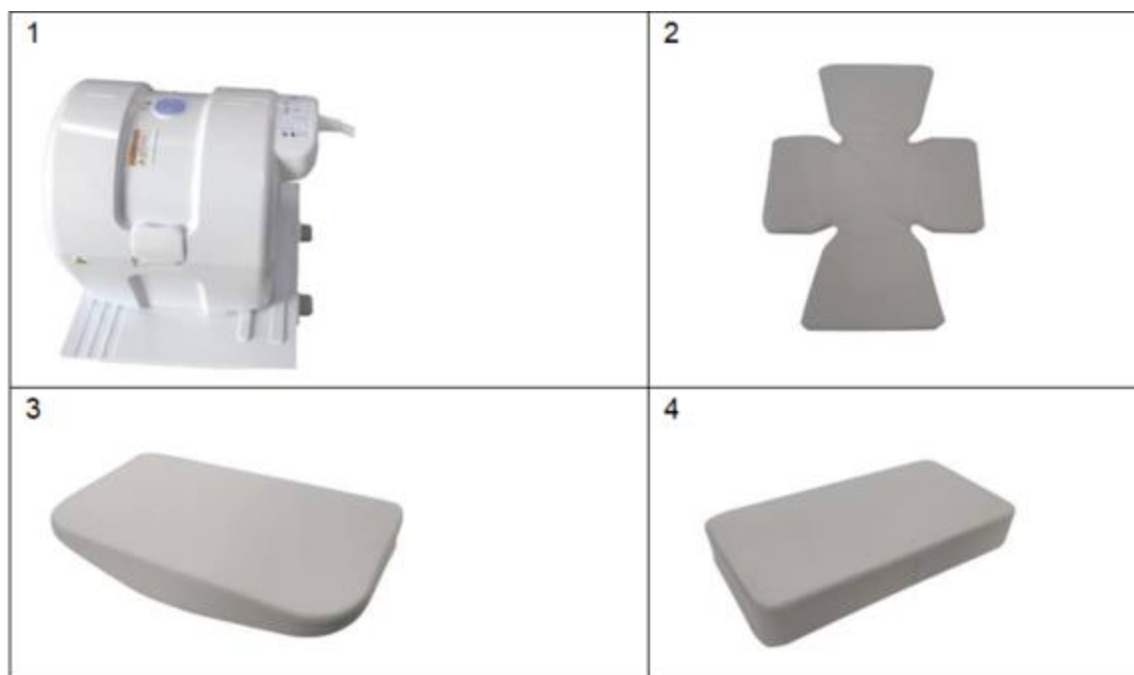


Table 1-5: Applied parts

Content			
NO.	Description	RFT part#	GE part#
1	8ch T/R Knee Coil	10-F31896	5869241-2
	1.5T HD T/R Knee Array	10-F34127	5932028-2
2	8ch T/R Knee Coil Inner Down Pad	6-F32413	5869241-4
3	8ch T/R Knee Coil Ramp Pad(2pcs)	6-F32054	5869241-5
4	8ch T/R Knee Coil Side Pad	6-F32055	5869241-6

NOTE: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR system.

1-3 Introduction

This manual describes the safety precautions, features, use and care of Knee Coil, for use with GE 1.5T MRI systems. Knee Coil is a transmit/receive coil. Please read this manual thoroughly before using the device.

If you have any questions or comments regarding this manual, or if you need any assistance with the use of this product, please contact an authorized manufacturer representative.

1-4 Compatibility

The connector of Knee Coil is P-Port or A-Port, connect to the specific port with T/R terminal of the GE system, please note that the P-Port or A-Port plug tag. The Knee Coil with P-Port is compatible with GE 1.5T **SIGNA Victor & SIGNA Prime** where coil ID allows. The Knee Coil with A-Port is compatible with GE 1.5T **SIGNA Explorer** where coil ID allows. See the system documentation for details.



Illustration1-6 P-Port



Illustration1-7 A-Port

1-5 Related Documentation

Knee Coil Operator Manual, Direction 81-F31896.

1-6 Indications for Use

The Knee Coil manufactured by Shenzhen RF Tech Co.,Ltd is transmit/receive coil and is designed for use as general purpose coil. The Knee Coil is designed to be use with GE 1.5T MRI systems to produce diagnostic images of the knee, palm, sole, ankle, elbow, wrist that can be interpreted by a trained physician.

1-7 Qualification for Service Personnel

Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

1-8 Warranty Statement

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Any modification of this coil may affect compatibility. Prior to modifying the coil, please contact your GE Healthcare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

1-9 Cautions



No modification of this equipment is allowed.



Do not modify this equipment without authorization of the manufacturer.



If this equipment is modified, appropriate inspection and testing must be conducted to ensure continued safe use of equipment.



Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

1-10 Coil Classification

In accordance with the international safety standard IEC 60601-1, this system is a Class II device, acceptable for Continuous Operation, having ordinary protection against ingress of water with type BF applied parts and is not for use in the presence of flammable anesthetics.

1-11 Environmental Requirements

This equipment must be stored under the following conditions:

- Ambient temperature range of -30°C to $+70^{\circ}\text{C}$.

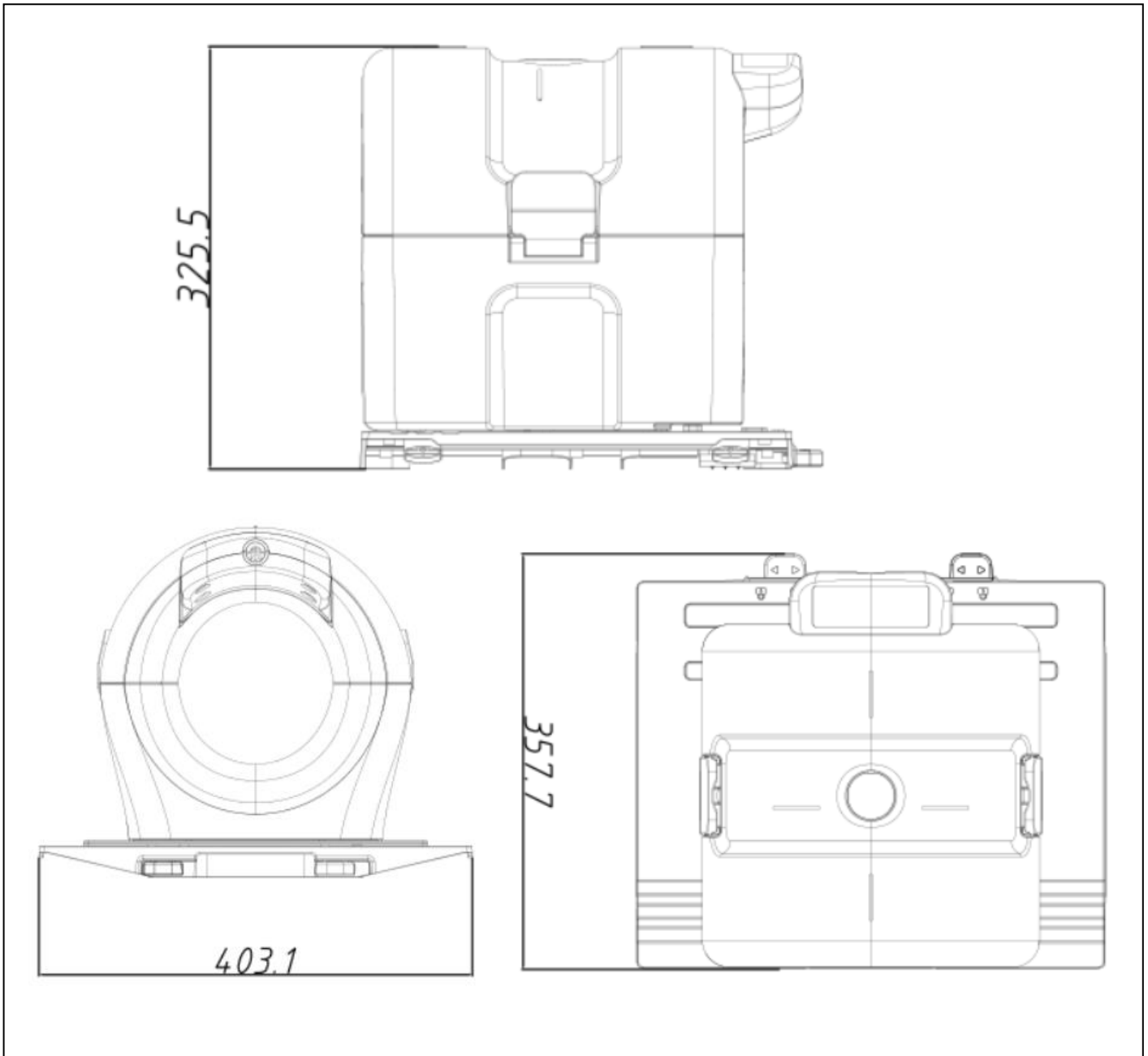
- Relative humidity range of 5% to 95%, non-condensing.
- Atmospheric pressure range of 500 hPa to 1,060 hPa.

Table 1-8: Dimension

Component	Coil	
	Dimensions (L x W x H)	Weight
Knee Coil	40.4cm x 35.8cm x 32.6cm	6.1Kg

The images below show more detailed dimensions of the Knee Coil.

Illustration 1-9: Knee Coil Dimension



Chapter 2 Coil Setup

2-1 Knee Coil MCQA Test Setup

2-1-1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	0 min	15 mins	0 min

2-1-2 Overview

Quality Assurance testing should be performed upon receipt of the coil to establish a baseline of coil performance. Perform system level Signal to Noise Check. The Quality Assurance testing frequency, please refer to the frequency of system service.

NOTE: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR system.

2-1-3 Preliminary Requirements

a. Tools and Test Equipment

Test Coil	Phantom	Quantity	GE part #
Knee Coil	Large, Cylindrical Unified Phantom	1	5342679

b. Safety



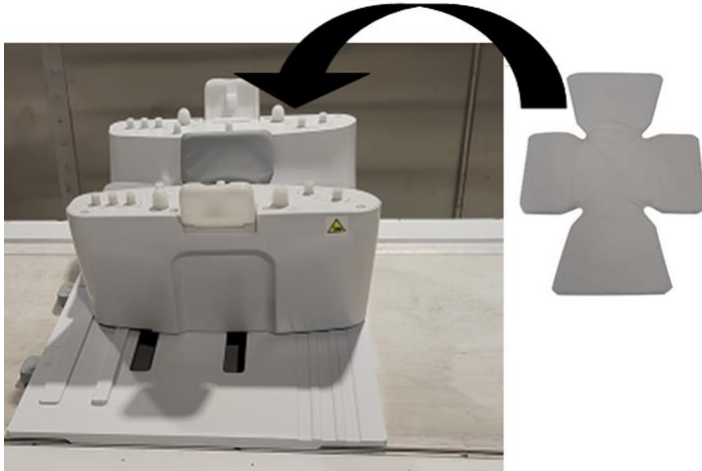
WARNING

ALL RF COIL-RELATED TESTS MUST BE RUN ON A SYSTEM THAT IS WELL CALIBRATED AND PASSES ALL SYSTEM TESTS. THE SYSTEM SHOULD HAVE PASSED INSTALL IN SPEC (IIS), ESPECIALLY WHITE PIXEL, CORRELATED NOISE, AND MULTI-COIL-RECEIVE (MCR) TOOL.

2-1-4 Procedure

1. Place the lower section of the coil on the patient table.
2. Put the Pad on the coil lower base.

Illustration 2-1: Place the pad



3. Place the Large Cylindrical Unified phantom into Coil.

Illustration 2-2: Place phantom



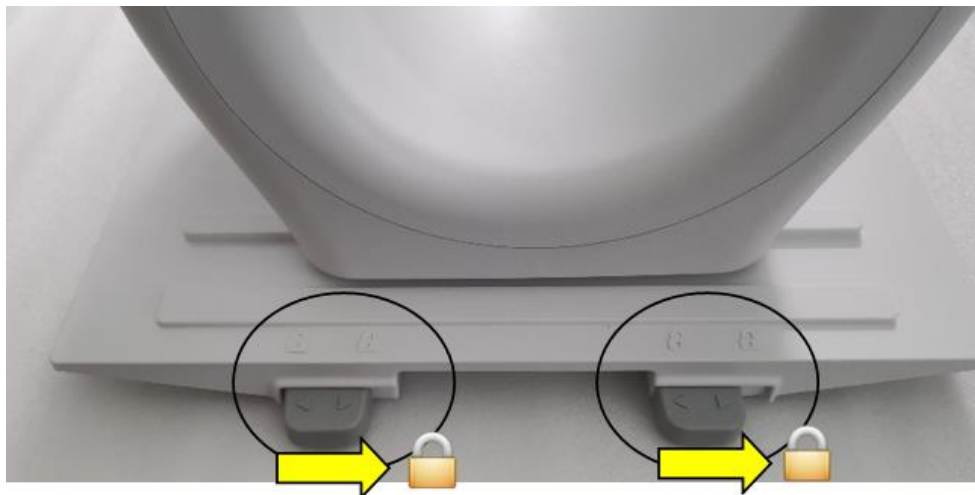
4. Carefully align the connector pins on the lower part of the coil with the upper part of the coil and latch both the coil sections into place. (NOTE: The scanner will not operate if the coil sections are not latched correctly.)

Illustration 2-3: Latch clip



5. Lock the coil in the desired position.

Illustration 2-4: Lock button



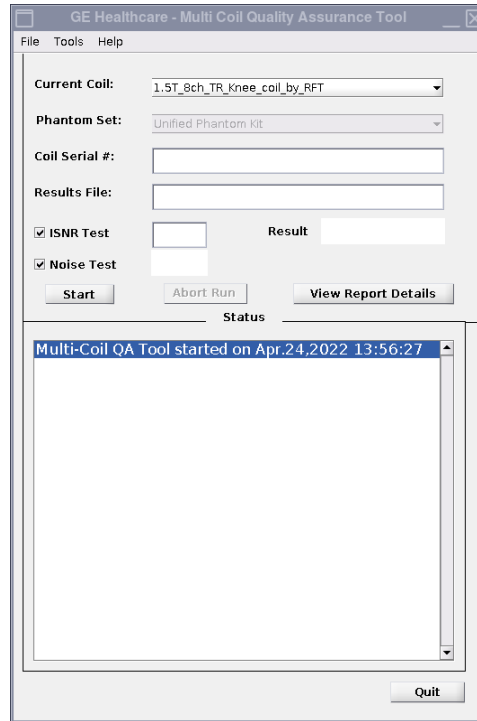
6. Connect the coil to P2 port (P-Port connect use) or A port (A-Port connect use).
7. Landmark to the cross mark.

Illustration 2-5: Landmark the coil



8. Run the MCQA check. (NOTE: Select 1.5T_8Ch_TR_Knee_coil_by_RFT in the Current Coil Field.)

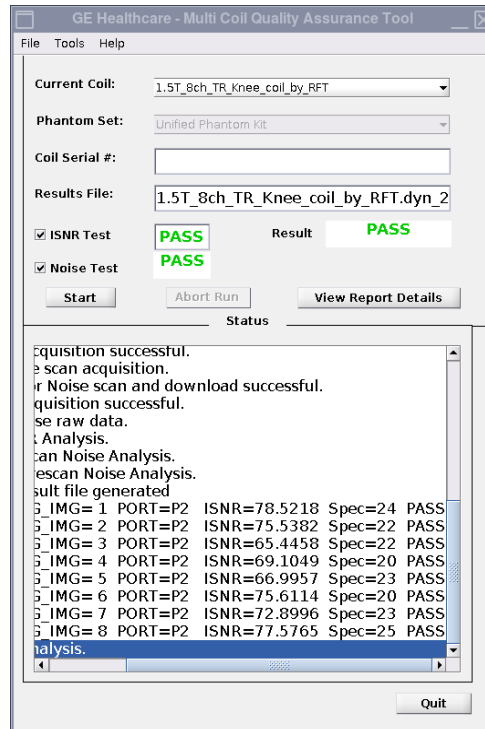
Illustration 2-5: MCQA tool menu



2-1-5 Data Analysis

When the scan completes the data will be displayed in the Multi-Coil Quality Assurance Tool Control window (illustration 2-6). Each channel along with the composite image is assigned pass/fail value.

Illustration 2-6: Test Results



Chapter 3 Coil Use

3-1 Knee Coil Use

3-1-1 Base Positioning



Figure 3-1

The Knee Coil is equipped with a fixed base. The base is directly placed on the patient's scanning table to ensure that the coil is stable.

For knee imaging, orient the coil bore toward the magnet end of the patient table to image the patient's knee in the supine position.

3-1-2 Coil Positioning Left and Right Positioning of Coil



Figure 3-2

Instructions for base switch:

 It means that the base switch is unlocked, and the knee coil can be slid;

 It means that the base switch is locked, the knee coil is fixed, and cannot be slid;

The operation process is as follows in actual use:

Slide left and right and turn the button at , so that the knee coil slides left and right on the base to adapt to the knee examination on the left or right.

The coil in the figure is located on the left knee.

Slide the knee coil to the desired position. Then rotate the latch to position  where the coil is locked to the base.



Pay attention to the slot during movement to prevent pinching.

3-1-3 Cushion Configuration



Knee Coil is equipped with 4 cushions.

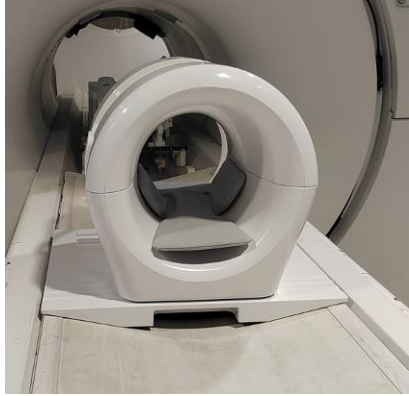


Figure 3-3

The cavity cushion is used to center the knee in the coil. It also ensures patient comfort.

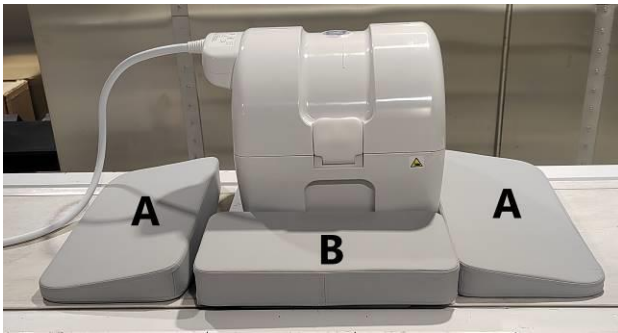


Figure 3-4

Ramp cushion [A] supports the scanned calf or ankle to ensure patient comfort.

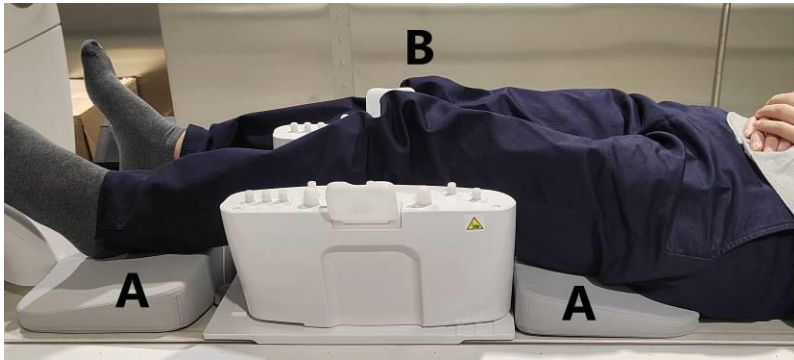


Figure 3-5

Side cushion [B] ensures patient comfort and supports the non-imaged knee to reduce motion of the knee not imaged during scanning.

3-1-4 Patient Knee Positioning



Figure 3-6

Knee scanning:

Let the patient lie flat on the table with one knee imaged in the coil, so that the apex of the patella is centered in the coil and aligned with the positioning line.

3-1-5 Patient Elbow/Wrist Positioning



Figure 3-7

Elbow scanning:

The patient extends the arms forward in a prone position, with head first entering the scanning environment, and then the patient lies on the bed with the face down, and centers the elbow to be scanned in the coil, aligned with the positioning line.

Wrist scanning:

The patient extends the arms forward in a prone position, with head first entering the scanning environment, and then the patient lies on the bed with the face down, and centers the wrist to be scanned in the coil, aligned with the positioning line.

3-1-6 Coil Locking

Align the connector with care, place the upper portion of the knee coil on the lower portion of the knee coil, and make sure that no foreign material obstructs the connection of the connector. Lock the upper and lower parts of the knee coil by locking the buttons on both sides of the upper portion of the knee coil. To open the coil, just open the buckles on both sides of the coil and lift the upper portion of the knee coil.



The scanner will not operate if connectors of the upper and lower parts of the knee coil are not properly locked.



Be sure not to pinch the patient when locking the upper portion of the knee coil to the lower portion of the knee coil.



Open the coil: Open the buckle on both sides and lift the



Lock the coil: Lock the buckle on both

3-1-7 Patient's Hearing



Provide ear plugs for the patient after the patient is informed of all instructions.



Hearing protection is required for all personnel in the scanning room during scanning, to prevent hearing impairment.



Acoustic levels may exceed 99 dB(A). Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB.)

3-1-8 Connect Coil to System



The connector for Knee Coil is P-Port or A-Port Connector, which is connected to the specific port on the GE system that contains a T/R terminal. Please note the P-Port or A-Port plug label. Connecting the wrong port will make it impossible to scan.



Do not cross or loop the cable. Failure to do so may result in arcing and burns to the patient.



The coil must be removed before lowering the patient table.



Lead the cable from the direction toward the magnet to avoid contacting with the patient.



Figure 3-8 P-Port



Figure 3-9 P-Port

3-1-9 Coil Positioning

The patient is fed into the magnet for positioning on the positioning line, as shown in the figure below.

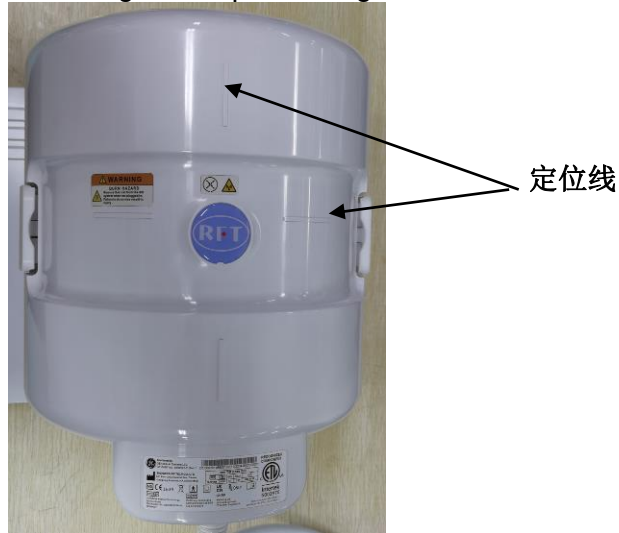


Figure 3-9 Coil Positioning



Once the scanning is completed, the coil must be removed before lowering the patient table.

Chapter 4 Scanning

4-1 Autoshim

Generally, the image quality can be improved by enabling autoshim. AUTOSHIM is a feature of the GE MRI System to improve image quality. It does this by improving the magnetic field homogeneity within the FOV selected. When the selected FOV is far from the center, autoshim can improve the quality significantly.

4-2 Positioning

The 8-Channel T/R Knee Coil is designed in such a manner that the body coil is not allowed to be used when the coil is connected to the scanner.

4-3 Fat Saturation Techniques

Off-center FOV imaging is a more complex and difficult technique since it is dependent upon the homogeneity of the magnetic field and the determined fat peak signal. Imaging at the edge of the magnet may produce poor fat saturation.

For best fat saturation results, position the patient as close to isocenter as possible, while ensuring that the non-imaged foot is away from the coil to avoid phase overlapping.

For axial imaging, use an axial positioner, but before specifying the slices, use the same FOV you intend to use in your study for positioning or use explicit positioning. Use of graphic indication from a large FOV positioner sometimes results in software error, thus producing blank slices, shifted slices, or both.

4-4 Scanning Protocols

GE Corporation recommends that you select imaging protocols that have been created by your radiologist. In addition, you may refer to the GE protocols acting on the system.

Chapter 5 Troubleshooting

5-1 Knee Coil Troubleshooting Tips

5-1-1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	0 min	45 mins	0 min

5-1-2 Overview

The following tips can be used to troubleshoot common problems with the Knee Coil.

5-1-3 Preliminary Requirements

a. Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Digital Volt Meter	1	-	-	-

5-1-4 Procedure

a. Receiving No Signal

• Problem:

You are unable to pre-scan or are scanning and yet receiving no signal.

• Possible Solution:

- Verify that you have made the appropriate system coil selection. Refer to the Operators Manual for additional information.
- Verify that the landmark is correct and that the cradle has not unlatched.
- Verify that the scan locations and any FOV offsets are correct.
- Perform a continuity check on the external cable (below). The continuity check must be performed by a GE authorized Service Engineer.
- Verify that the coil is properly positioned.
- If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil.

If you still receive no signal, the problem probably lies with your MR system. If the scan completes successfully, there is probably a problem with the coil. Contact GE for further assistance. If you are unable to scan with the substitute coil, there may be a system problem related to this particular coil type.

b. Image Quality

- Problem:

Image quality is not what you expect.

- Possible Solution:

- Review the selected protocol. If you are performing the Periodic Quality Assurance, be sure your protocol is identical to the protocol provided in your Systems Recommended Protocols. If you are performing diagnostic images, you may need to increase NEX or FOV.
- Perform a continuity check on the external cable (below). The continuity check must be performed by a GE authorized Service Engineer.
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.
- Verify that your center frequency is within the frequency adjustment range for your system.
- Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.
- Perform a MCQA (Multi Coil Quality Assurance) Test. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using both another coil of the same type plugged into the same system port. If the image quality is visibly improved, there may be a problem with the coil. Contact GE for further assistance. If the image quality still suffers, there may be a system problem related to image with this type of coil.

c. Artifacts

• Problem 1:

There is a black line or signal void on the image.

• Possible Solution for problem1:

Verify that there is no metal present in the area being scanned in or on the patient.

• Problem2:

Some or all of the images appear shaded or exhibit uneven signal or banding.

• Possible Solution for problem2:

Confirm that no metallic objects are located nearby, outside the FOV. This is especially important on images utilizing Fat Saturation. If Fat Saturation is being used, verify that the Center Frequency fine adjustment has been optimized.

d. External Cable Wear

• Problem:

The system cannot recognize the coil or scan with the coil attached.

• Possible Solution:

Perform the cable continuity check. As follows:

RF Signal Channels Check: The center pins of the System Connector receive coaxial connectors are extremely fragile. Do not attempt to make any measurements at these center pins.

NOTE: With the coil cable disconnected, use a multimeter to assure an open circuit condition exists between the designated pin and GND for each signal channel shown in Table 5–1.

Table 5-1: Coil Cable Connector Short Circuit and Open Circuit Check

Signal Channel	Coil side connector RF GND Shield	Coil side connector RF Center PIN
CH1	S1	R1(C1)
CH2	S2	R2(C2)
CH3	S3	R3(C3)
CH4	S4	R4(C4)
CH5	S5	R5(D1)
CH6	S6	R6(D2)
CH7	S7	R7(D3)
CH8	S8	R8(D4)
TX	TX	T1(F2)
RX	RX	T2(H3)

- If one or more of the readings indicate a short, replace the cable.

Table 5-2: PINS and Connected DC wire

Pin in P-Port	5	6	18
Wire color	White10-Center Pin	White10-GND	White9

Pin in A-Port	M7	M8	M3
Wire color	White10-Center Pin	White10-GND	White9

- There are three pins (see *Illustration 5-2*) connecting to DC wires of the cable. *Table 5-2* shows these pins and corresponding DC wires. Use a Digital Multi-meter(DMM) to check short/open between the pin and the corresponding DC wire (see *Illustration 5-2*). The display should be short circuit. If DMM reading is open circuit , replace the cable.

Illustration 5-2: Coil Cable P-Port Connector

Inside of Interface	System Connector	Diagram
---------------------	------------------	---------

DC WIRES	PIN
White9	18
White10-Center Pin	5
White10-GND	6

RF WIRES	PIN
Short Circuit	1 and 2
White1	R-1
White2	R-2
White3	R-3
White4	R-4
White5	R-5
White6	R-6
White7	R-7
White8	R-8
Black	T1
Brown	T2

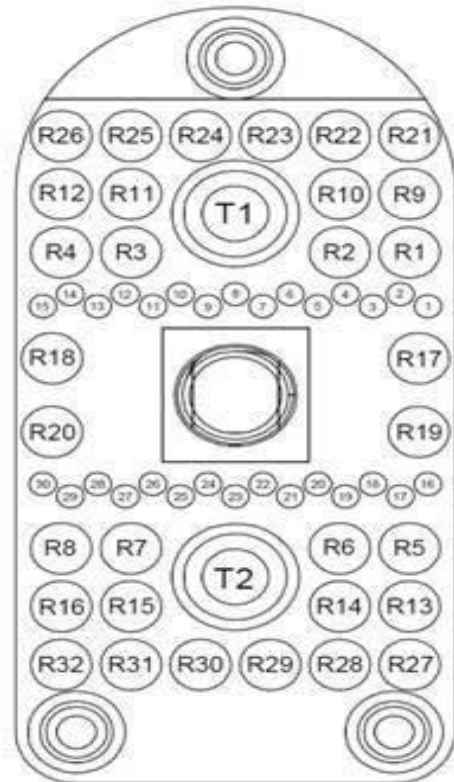
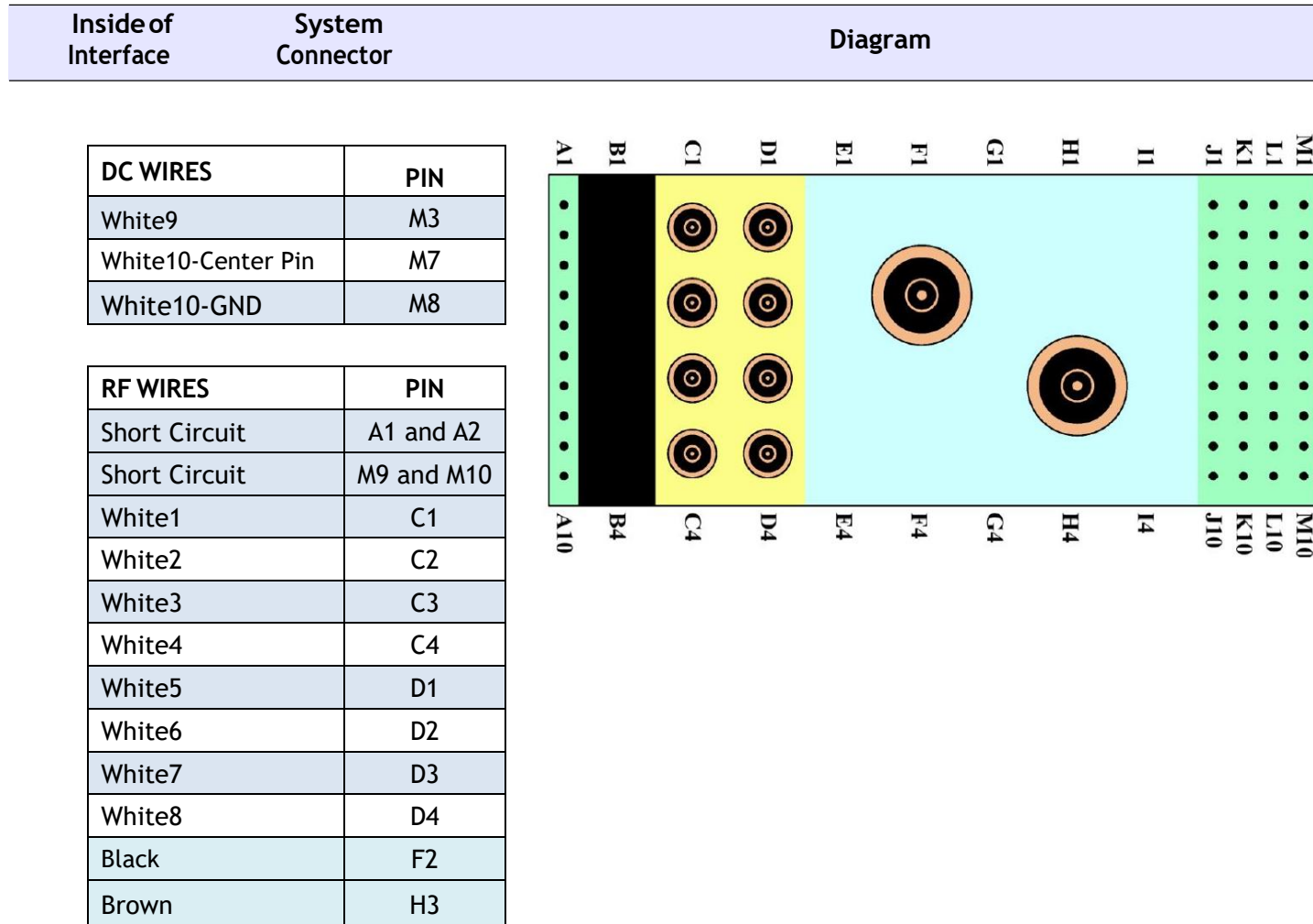


Illustration 5-3: Coil Cable A-Port Connector



e. Mechanical Parts Wear

Inspect the coil for visible signs of wear. If the coil housings cannot seat with each other or if the latch mechanism cannot work properly, contact Service.

5-1-5 Finalization

No finalization steps.

Chapter 6 Maintenance

6-1 Quality Assurance

The procedure described in the Quality Assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals according to the requirements of MR system Operator Manual.

6-2 Coil Care

WARNING!

Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.

CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: a ten percent bleach solution (some discoloration may occur) or warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

6-3 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

Chapter 7 FRU and Replace procedures

7-1 Knee Coil FRUs

The following tables list the Field Replaceable Units (FRUs) and other replaceable parts for the Knee Coil.

7-1-1 Field Replaceable Units

Table 7-1: Field Replaceable Units

Description	GEHC part number
8ch T/R Knee Coil w/cable and baseplate	5869241-2
1.5T HD T/R Knee Array w/cable and baseplate	5932028-2
8ch T/R Knee Coil Cable	5869241-3
1.5T HD T/R Knee Array	5932028-3
8ch T/R Knee Coil Inner Down Pad	5869241-4
8ch T/R Knee Coil Ramp Pad	5869241-5
8ch T/R Knee Coil Side Pad	5869241-6

7-2 Knee Coil Cable Replacement

7-2-1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	0 min	30 mins	0 min

7-2-2 Preliminary Requirements

a. Safety



NOTICE

This procedure must be performed by an authorized GE Service Engineer only.

7-2-3 Procedure

a. Cable Replacement

1. Proceed as follows for the Cable Replacement procedure:

Step 1. Remove all screws on the upper coil cable port cover and remove the coil cover .

Step 2. Remove the four brass screws that hold down the cable clamps(see illustration 7-1).

Step 3. Disconnect the cable by grasping the coil connector housing, and pulling the cable away from the coil connector in a downward direction ((see illustration 7-2)) .

Step 4. Remove the old cable assembly.

Step 5. Fix a new cable assembly on the T/R knee coil and reattach the cable lock with four brass screws.

Step 6. Connect the coil connector housing to the corresponding receptacles on the interfaces board.

Step 7. Close the coil cover lock with 4 RENY screws.

Step 8. Perform MCQA after replacement.

Illustration 7-1 Unfastening the cable from the T/R Knee Coil

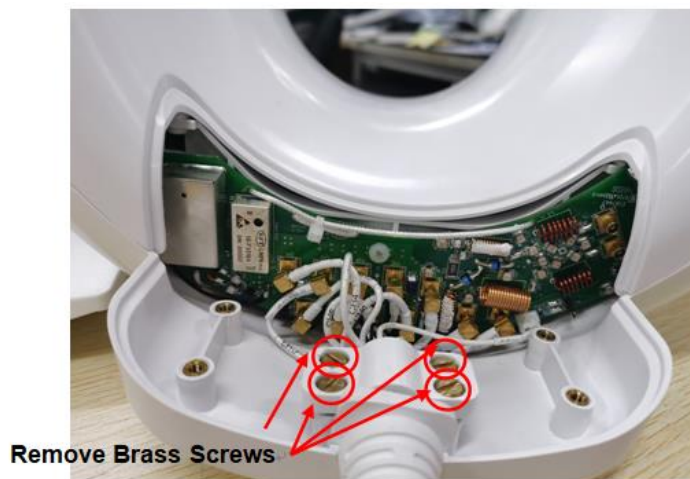


Illustration 7-2 Unfastening the cable from the T/R Knee Coil



7-2-4 Finalization

No finalization steps.

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